

Pictures Made of Splats: Turning Pages into Clouds of Gaussians

The Splashery Lab

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Abstract

We show how a printed page, a photo or a frame of video can be rebuilt as thousands of small, flat Gaussian splats, and viewed from any side like a sheet of paper. The paper becomes a smooth sheet of large splats in its own color; every pixel of ink becomes one small splat of its own color, a hair in front of the paper. Pages are built when they are reached and freed when they are left, so a book of any length costs no more than its open pages. We measure how many splats a page needs to stay readable on a phone, and how the detail should follow the view.

1. Introduction

Gaussian splatting draws a scene as a cloud of soft, colored ellipsoids. Each splat has a center, a size along three axes, a turn, a color and an opacity, and the renderer sorts them by depth and blends them from back to front. Scanned objects often use a million splats or more. A page of text is a different kind of subject: it is flat, it has sharp edges, and most of it is plain paper.

A splat is not a pixel. Its edge is soft, and splats that overlap blend their colors. Laid out on a grid with the right size, though, small flat splats close up into a surface with no gaps, and the eye reads them as the picture they came from. This article describes one simple way to do that for documents and photos, and what it costs.

2. From pixels to splats

For a photo, every pixel becomes a flat disc facing the viewer. Its standard deviation is 0.6 of the pixel's width, so neighboring discs close up to about ninety-six percent cover without blurring the picture by more than a pixel. A smooth base of larger splats, a hair behind, fills what is left and keeps the back of the sheet solid.

$$G(x) = \alpha \exp(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu))$$

For a document, most pixels are paper. We find the paper's color block by block, from the lighter half of each block, so an off-white or

unevenly lit scan works too. Only the pixels that differ from the paper under them, the ink, get their own splats.

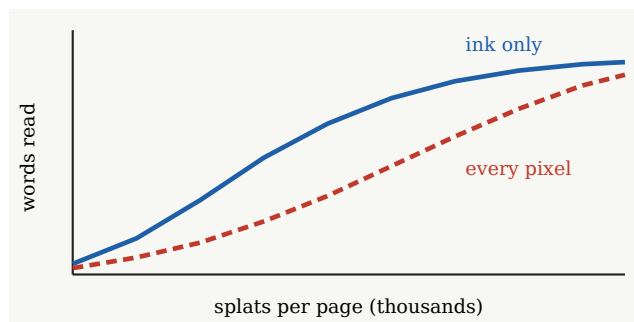


Figure 1. Words read on a phone against the splats a page uses. Keeping only the ink reads well with far fewer splats.

3. Near and far

A renderer skips splats smaller than about a pixel on the screen. A whole page seen on a phone is about eight hundred pixels wide, so a page built at print resolution would vanish there, while a page built for the phone would look soft when zoomed in. We build each page at about one of its pixels per screen pixel for the size it shows at now, and rebuild it when the view comes much closer or goes much further.

Tier	Page width	Ink splats	Photo splats
Low	1,100 px	220,000	160,000
Mid	1,600 px	420,000	320,000
High	2,400 px	750,000	640,000
Max	3,200 px	1,300,000	1,200,000

4. Pages that stream

A book never builds all its pages. Only the open pages and the one turning exist as splats. A page is drawn from the file when it is reached, turned into splats on a worker thread in a fraction of a second, and freed when it leaves. The splat count stays the same whether the book has ten pages or two hundred.

Video works the other way round. Its splats are built once, and each one takes its color from a small picture of the current frame, which the graphics card receives thirty times a second.

5. Results

On a phone, the title and headings of a page read at a glance with the whole page in view, and every word of the body text reads when zoomed in. Colors match the source to within the rounding of half-precision numbers, and the paper shows no grid.

The sheets bend like paper. Each splat of a page knows how far it lies from the book's spine; as a page turns, it curls along a circular arc and every splat turns with the curve, so the page stays solid at every angle.

6. Conclusion

Pictures and pages make good splat toys. The same small set of rules covers a research article, a family photo and a home video, and it runs in a browser on a phone. Fitting splats of free shape to a picture, as recent work does on large graphics cards, could make pages lighter still.

References

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